L2.2 The Mean Value Theorem

§3.2 — what you need to know

The six theorems

①	**IVT**	A continuous curve sweeps out all y-values between its minimum and maximum on some interval.
②	**EVT**	On a closed interval, a continuous function reaches a highest and a lowest value.
③	**Fermat**	At a local max or min where f' exists, $f'(c) = 0$.
④	**Closed Interval Method**	Evaluate f at the critical numbers and both endpoints; take the largest and smallest.
⑤	**Rolle**	Equal endpoint heights force a horizontal tangent between them.
⑥	**MVT**	Somewhere between the endpoints, the tangent is parallel to the secant.

To use each theorem, you must show that any needed hypotheses are true. Otherwise, you aren't able to use the theorem.

ROLLE'S THEOREM AND THE MEAN VALUE THEOREM

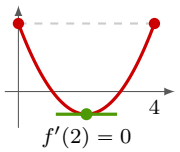
Rolle. If f satisfies:
1) continuous on $[a, b]$
2) differentiable on (a, b)
3) $f(a) = f(b)$

$\exists\, c$ on (a, b) s.t. $f'(c) = 0$

Ex. Verify that $f(x) = x^2 - 4x + 3$ satisfies the hypotheses of Rolle's Theorem on $[0, 4]$, then find c.

1) Polynomial, so cont on $[0, 4]$ ✓
2) Polynomial, so diff on $(0, 4)$ ✓
3) $f(0) = 3$ and $f(4) = 16 - 16 + 3 = 3$ ✓

Rolle: $f'(c) = 2c - 4 = 0 \;\Rightarrow\; c = 2$



MVT. If f satisfies:
1) continuous on $[a, b]$
2) differentiable on (a, b)

$\exists\, c$ on (a, b) s.t. $f'(c) = m_{\text{sec}}$

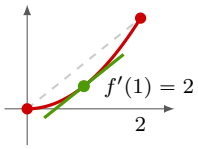
$m_{\text{sec}} = \dfrac{f(b) - f(a)}{b - a}$, the slope of the secant through the endpoints.

Ex. Verify that $f(x) = x^2$ satisfies the hypotheses of the Mean Value Theorem on $[0, 2]$, then find c.

1) Polynomial, so cont on $[0, 2]$ ✓
2) Polynomial, so diff on $(0, 2)$ ✓

MVT: $m_{\text{sec}} = \dfrac{4 - 0}{2 - 0} = 2$

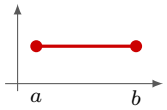
$f'(c) = 2c = 2 \;\Rightarrow\; c = 1$



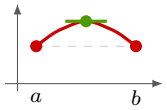
Rolle is the case $m_{\text{sec}} = 0$. Dashed: the secant. Green: the tangent at c.

DERIVATION OF ROLLE'S THEOREM

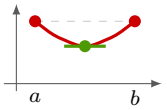
There are only three possibilities if $f(a) = f(b)$: $f(x)$ stays constant, it goes up in between, or it goes down.



① f **constant.** $f' = 0$ everywhere on (a, b) — any c works.



② f **goes up. EVT** gives a Max; the equal ends put it *inside*; **Fermat** gives $f'(c) = 0$.



③ f **goes down.** The same argument with a min.

Case ② uses both hypotheses: EVT needs f continuous on $[a, b]$, and Fermat needs f differentiable on (a, b).

PROOF BY CONTRADICTION

"assume something you don't know is true and see what happens — if it leads to a contradiction, the thing you assumed is wrong"

The template, and it works on any *at most one* claim:

① Assume the **opposite** of what you want. Suppose you want to show that only one root exists, and you already know at least one does: assume the opposite, e.g., that two roots exist.

② Apply a theorem whose hypotheses are true because of the assumption.

③ Push to a statement that contradicts something already known — usually a fact about f'.

④ The assumption was false.

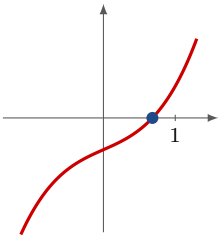
EXACTLY ONE REAL ROOT

Ex. Show $x^3 + x - 1 = 0$ has exactly one real root.

Let $f(x) = x^3 + x - 1$.

Step 1) At least one root exists, via IVT.

1) Polynomial, so cont on $[0, 1]$ ✓
2) $f(0) = -1$ and $f(1) = 1$, so 0 is between ✓
IVT: $\exists\, c$ on $(0, 1)$ s.t. $f(c) = 0$.



(Just for visualization. The proof has to come from IVT and Rolle, not from the graph.)

Step 2) Only one root exists.

Assume, for the sake of a contradiction, that two roots exist, $a < b$.
Then $f(a) = f(b) = 0$. (both are roots, so same y value)
1) cont on $[a, b]$ ✓ 2) diff on (a, b) ✓ 3) $f(a) = f(b)$ ✓
Rolle: $\exists\, c$ on (a, b) s.t. $f'(c) = 0$.
But $f'(x) = 3x^2 + 1 \geqslant 1$, so f' is never 0.
Contradiction, so there is no second root.
Exactly one real root.